Eduardo de Jesús Dávila Meza, Ph.D.

AI/ML & DL • Computer Vision • Embedded/Robotics Systems (ROS/ROS2)

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Federal Professional Certificates: Bachelor's Degree: 12027207 • Master's Degree: 14043743 • Ph.D. Degree: 15067339

Engineering Products

Vision Pipeline – Production-oriented Computer Vision Inference Framework | Modular FastAPI/Docker inference system using YOLO11 and PyTorch, with REST/OpenAPI interfaces, YAML configuration, structured logging, CLI inference, JSON response schema, automated testing, CI, and reproducible benchmarking.

Python · PyTorch · YOLO11 · FastAPI · Docker · OpenAPI · pytest · GitHub Actions

ROS/ROS2 Courses – Robotics Software & Perception Systems | Modular ROS/ROS2 repositories covering differential-drive-robot programming, simulation, computer vision, navigation, embedded Linux and deployment-oriented robotics workflows.

Stack: C++ · Python · ROS/ROS2 · OpenCV · Linux · Gazebo · RViz · Bash

Technical Skills

AI/Computer Vision: Computer Vision, Deep Learning (Mask R-CNN), Object Detection, Image Segmentation, Model Optimization, PyTorch, TensorFlow/Keras, OpenCV
Software Engineering: Python, C++, SQL, Bash, REST APIs, FastAPI, OpenAPI, pytest, JSON, XML, YAML
Deployment/Infrastructure: Docker, Docker Compose, Git, GitLab, GitHub, GitHub Actions
Robotics/Embedded: ROS/ROS2, RViz, Gazebo, Embedded Linux, C++/Python robotics development
Data/ML: NumPy, Pandas, Scikit-learn, Matplotlib, SQLAlchemy, Time Series Analysis (ARIMA), LSTM
GNU/Linux: Ubuntu & Arch Linux
Documentation: LaTeX, Markdown, MediaWiki, MS Office, Dia Diagram

Engineering Focus

- Computer vision and deep-learning systems
- Modular AI inference and deployment architectures
- ROS/ROS2 perception and robotics systems
- Production-oriented Python/C++ development
- API, testing and reproducible deployment workflows

Spoken Languages

Spanish | Native: full professional proficiency.
English | B2 Advanced / Professional Working Proficiency. Certified by Cinvestav and Desafío Latam.

Work Experience & Projects | Full List

Robot Programming Instructor – ROS/ROS2 — **April 2024 – Present**
Tecnológico de Monterrey (ITESM), Guadalajara — *Zapopan, Jalisco*
1) Designed and delivered hands-on courses in ROS/ROS2 using C++ and Python, focused on differential drive robots. **2)** Developed practical repositories and lecture notes covering the foundations of development, simulation, deployment, and monitoring of ROS/ROS2-based modular systems, including computer vision and autonomous navigation. **3)** Recognized as a top-rated instructor (See recognition). See repository 2025. See latest repository.

C++ · Python · ROS/ROS2 · Bash · Computer Vision · Robotics · Embedded Linux · Technical instruction

Mathematical Analyst in Information Technologies — **September 2025 – April 2026**
Geovoy, Busmen Group — *Tlajomulco de Zúñiga, Jalisco*
1) Built an automated reporting tool using LLM-powered APIs to generate analytical reports and insights. **2)** Developed statistical and machine learning models (time series analysis (ARIMA), LSTM) for route optimization and demand prediction, generating insights to improve operational decision-making. **3)** Designed and optimized APIs using FastAPI, raw SQL, and Object-Relational Mapping (ORM) to integrate statistical models into production systems for geolocation and inventory management. **4)** Deployed a MediaWiki-based platform to standardize internal processes, code, and API documentation.

FastAPI · SQLAlchemy · Time Series · LSTM · Forecasting · Predictive Modeling · GPT-OSS-20B · Ollama · Data Analysis · Bash

AI/Deep Learning Research Engineer – Computer Vision — **May 2021 – July 2024**
Cinvestav, Guadalajara: Medical Image Segmentation Project — *Zapopan, Jalisco*
1) Developed a deep learning model for medical image analysis, in collaboration with German research institutions, to support clinical decision-making, focusing on segmentation and quantification of retinal pathologies. **2)** Applied data preprocessing, augmentation, feature extraction, and model validation techniques to ensure robustness and reliability in healthcare-related datasets. **3)** Worked with high-resolution image datasets to improve the detection and segmentation of retinal pathologies. **4)** Contributed to a research publication in computer vision and medical imaging.

Computer Vision · Deep Learning · TensorFlow/Keras · Mask R-CNN · Image Segmentation · Data labeling, augmentation, and visualization

Computer Vision & ROS Developer — **December 2019 – October 2023**
Cinvestav, Guadalajara: Intelligent Visual Guide System (OJO SMART) – Modular Navigation Device — *Zapopan, Jalisco*
1) Developed ROS-based modules for real-time recognition of colors, objects, signs, banknotes, and text using computer vision techniques. **2)** Contributed to the integration of multiple perception systems into an assistive navigation device for visually impaired users.

Computer Vision · C++ · Python · ROS · Tesseract OCR · OpenCV · TensorFlow · Embedded Linux

Publications & Patents

Meeting Abstract | **June 2024** | "Deep-learning based quantification of RPE65-mutation inherited retinal degeneration", presented at *Investigative Ophthalmology & Visual Science*, vol. 65(7), 1392, ID: 2794864.

AI/ML lifecycle · Computer vision / Medical image analysis · Data visualization · Mask R-CNN · Feature extraction · Research

Journal Article | **September 2023** | "Quaternion and Split Quaternion Neural Networks for Low-Light Color Image Enhancement", in *IEEE Access*, vol. 11, 108257-108280, 10.1109/ACCESS.2023.3312234.

AI/ML lifecycle · Computer vision / Image color analysis · Quaternion algebras · Color spaces · EKF

Patent | **March 2017** | "Device for controlling underactuated two-link systems with one actuator", filed under the Invention Support Program, University of Guadalajara. Application no. MX/a/2017/016436.

Embedded systems · Control theory · Digital and power electronics · PICs · SPI & I2C communication protocols

Academic Degrees

Ph.D. in Science – AI/Deep Learning | **Cinvestav, Guadalajara** — **September 2019 – May 2024**
Thesis | Deep learning for recognition and quantification of fundus pathologies using instance segmentation, and quaternion neural networks for low-light image enhancement. Download thesis.

AI/ML lifecycle · Computer vision / Image analysis · Deep Learning · CNN/NN · Robotics · Research · Science Communication

M.Sc. – AI/Machine Learning | **Cinvestav, Guadalajara** — **September 2017 – August 2019**
Thesis | Quaternion neural networks for low-light image enhancement, and identification of an electromechanical system. Download thesis.

AI/ANNs · Computer vision · Linear algebra · Control theory · Robotics · Probability & statistics · Research · Science Communication

B.Eng. in Mechatronics – Embedded Systems | **University of Guadalajara** — **August 2012 – December 2016**
Social Service & Professional Internship | Developed embedded and mechatronic projects in an electronics and telecommunications laboratory.

Embedded systems · Linear algebra · Calculus · Control theory · Digital and power electronics · HMI · PICs & Arduino

Certifications | Additional Training

AI/DL	AI/ML	Data Cleaning	LLMs	NLP	OpenCV	Pandas
Python	R for DS	ROS	Scikit-learn	SQL	TF-Keras	Full list